23

Matrix Groups

Historical Note

In 1872, on taking up a chair at Erlangen, Felix Klein proposed that a geometry is not a set of axioms about points and lines but a group of transformations together with the quantities that group leaves alone. Euclidean geometry is the study of what rigid motions preserve, projective geometry of what projective maps preserve, and so on. Sophus Lie, Klein's friend and collaborator, spent the following decades on the continuous groups that carry the idea, and discovered that such a group is almost entirely determined by its behaviour near the identity, which is a linear object. Elie Cartan classified them. Hermann Weyl, in a 1939 book that gave them their name, established the classical groups as the working vocabulary of physics, and Emmy Noether had already shown in 1918 that every continuous symmetry of a physical law yields a conserved quantity. All of these groups are groups of matrices, and this chapter is about the ones the earlier chapters have already built.

Chapter 22 classified bilinear forms. This chapter takes the other half of Klein's programme and asks, for each form, which matrices preserve it. The answer in every case is a group, and those groups turn out to be the ones already met: the orthogonal matrices of Chapter 12, the unitary matrices that made the spectral theorem work, and the determinant-one matrices that Chapter 8 showed preserve volume.

23.1 Groups of matrices

Definition 23.1 (Matrix group)**.** *The invertible matrices $GL_n(\mathbb{F})$ form a group under multiplication, by Proposition 4.15: the product of invertibles is invertible, multiplication is associative, $\mathbf{1}$ is a unit and inverses exist. A* matrix group *is a subgroup of some $GL_n(\mathbb{F})$, that is a set of invertible matrices closed under products and inverses and containing $\mathbf{1}$.*

Example 23.2 (The standard cast). *Each is a matrix group, and each is defined by preserving something.*

(i) $SL_n(\mathbb{F}) = \{A : \det A = 1\}$, *closed under products by Theorem 8.12. By Chapter 8 it is the group preserving signed volume.*

(ii) $O_n = \{Q \in M_n(\mathbb{R}) : Q^tQ = \mathbf{1}\}$, *the orthogonal group, and* $SO_n = O_n \cap SL_n(\mathbb{R})$.

(iii) $U_n = \{U \in M_n(\mathbb{C}) : U^*U = \mathbf{1}\}$ *and* $SU_n = U_n \cap SL_n(\mathbb{C})$.

(iv) The permutation matrices, a finite group with $n!$ *elements, met in Chapter 8.*

(v) The upper triangular invertible matrices, and inside them the unipotent *ones with ones on the diagonal.*

Proposition 23.3 (Each classical group preserves a form). *Let* β *be a bilinear or sesquilinear form on* $\mathbb{F}^n$ *with matrix* M*, and let* $G_\beta = \{A : \beta(Ax, Ay) = \beta(x, y) \text{ for all } x, y\}$. *Then* G_β *is closed under products, and consists of the* A *with* $A^tMA = M$ *in the bilinear case and* $A^*MA = M$ *in the sesquilinear case. Taking* $M = \mathbf{1}$ *gives* O_n *and* U_n *respectively; taking* M *to be the alternating form of Theorem 22.14 with* $n = 2r$ *gives the* symplectic *group* $\mathrm{Sp}_{2r}(\mathbb{F})$.

Proof. The condition $\beta(Ax, Ay) = \beta(x, y)$ reads $(Ax)^tM(Ay) = x^tMy$ for all x, y, that is $x^t(A^tMA - M)y = 0$ for all x, y, which forces $A^tMA = M$ by taking x and y to run over a basis. Closure under products is immediate from the definition, and when M is invertible the condition forces $(\det A)^2 \det M = \det M$, so $\det A = \pm 1$ and A is invertible; then A^{-1} preserves β too. □

Key Idea

This is Klein's principle in one line. A form is a piece of geometry, and the matrices preserving it are the motions of that geometry. Chapter 22 showed that over $\mathbb{R}$ there are only a few forms up to congruence, classified by inertia in the symmetric case and by rank alone in the alternating case, so there are only a few classical groups. The reason O_n, U_n and Sp_{2r} appear everywhere is not that anyone chose them. It is that there was almost nothing else to choose.

23.2 The orthogonal group

Proposition 12.15 already established that $Q \in O_n$ exactly when Q preserves inner products, exactly when it preserves lengths, and that $\det Q = \pm 1$. The sign splits the group in two.

Proposition 23.4 (Two components). *SO_n is a subgroup of O_n of index two, and the other coset consists of the orthogonal matrices of determinant -1. No continuous path inside O_n joins the two, since* det *is continuous and takes only the values ± 1 on O_n.*

Proof. By Proposition 12.15 the determinant of an orthogonal matrix is ± 1, so det restricts to a map $O_n \to \{+1, -1\}$, and it is a group homomorphism by Theorem 8.12. Its kernel is SO_n by definition. It is onto, because $\operatorname{diag}(-1, 1, \dots, 1)$ is orthogonal with determinant -1. A homomorphism onto a group of order two has kernel of index two, and the nontrivial coset is the preimage of -1, which is the second claim.

For the last sentence, suppose γ were a continuous path in O_n from a matrix of determinant 1 to one of determinant -1. Then $\det \circ \gamma$ is a continuous real function on an interval taking the values 1 and -1, so by the intermediate value theorem it takes the value 0 somewhere, contradicting $\det = \pm 1$ throughout O_n. □

Theorem 23.5 (The plane). *Every element of SO_2 is a rotation $R_\theta = \left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, and every element of $O_2 \setminus SO_2$ is a reflection in a line through the origin. Moreover $R_\alpha R_\beta = R_{\alpha+\beta}$, so SO_2 is a circle as a group as well as a set.*

Proof. Let $Q \in O_2$ have first column (a, c), a unit vector, so $(a, c) = (\cos\theta, \sin\theta)$ for some θ. The second column is a unit vector orthogonal to it, so is $\pm(-\sin\theta, \cos\theta)$. The plus sign gives R_θ, with determinant $\cos^2\theta + \sin^2\theta = 1$; the minus sign gives determinant -1 and the matrix $\left(\begin{smallmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{smallmatrix}\right)$, which is symmetric with trace 0 and determinant -1, hence has eigenvalues 1 and -1 with orthogonal eigenvectors by Theorem 14.7: that is a reflection. The addition formula is the trigonometric addition formula, computed directly. □

Figure 23.1 draws the conclusion: the orthogonal group of the plane is two circles, and only the one through the identity is a subgroup, since the product of two reflections is a rotation.

Theorem 23.6 (Euler, 1776). *Every $Q \in SO_3$ fixes a line: there is a unit vector a with $Qa = a$. In a right-handed orthonormal basis whose first vector is a, the matrix of Q is $\left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{smallmatrix}\right)$, so Q is a rotation through some angle about the axis a.*

Proof. It suffices to show 1 is an eigenvalue, that is $\det(Q - \mathbf{1}) = 0$. Using $\det Q = 1$, $Q^t Q = \mathbf{1}$ and Proposition 8.13,

$$\det(Q-\mathbf{1}) = \det Q \det(Q-\mathbf{1}) = \det\big(Q^t\big) \det(Q-\mathbf{1}) = \det\big(Q^t(Q-\mathbf{1})\big) = \det\big(\mathbf{1}-Q^t\big) = \det(\mathbf{1}-Q),$$

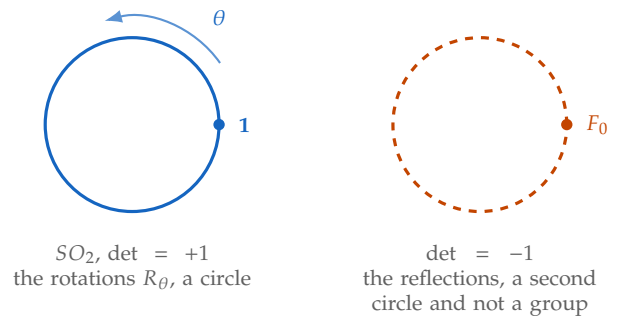


Figure 23.1. Theorem 23.5 and Proposition 23.4. The orthogonal group of the plane is two circles. Only the left one is a subgroup: the product of two reflections is a rotation, which is the two-dimensional case of the general fact that determinant is multiplicative. The same picture holds in every dimension, with SO_n connected and O_n having exactly two pieces.

while $\det(\mathbf{1} - Q) = (-1)^3 \det(Q - \mathbf{1}) = -\det(Q - \mathbf{1})$. Hence $\det(Q - \mathbf{1}) = 0$ and there is a unit a with $Qa = a$.

Now let $W = a^\perp$, which is Q-invariant: if $\langle w, a\rangle = 0$ then $\langle Qw, a\rangle = \langle Qw, Qa\rangle = \langle w, a\rangle = 0$. The restriction of Q to W is orthogonal with determinant $\det Q/1 = 1$, so lies in SO_2, and Theorem 23.5 makes it a rotation. □

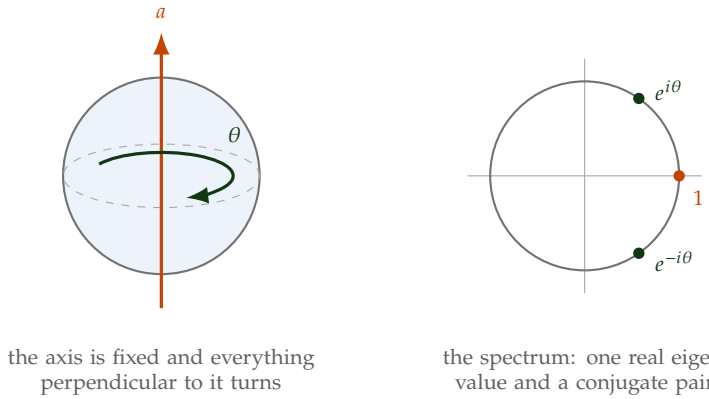
Figure 23.2 is the statement that is not obvious: two rotations of space about different axes compose to a rotation about a third. The eigenvalues on the right of it are forced, since Q preserves lengths and has real entries and determinant one.

23.3 The exponential, and the tangent space at the identity

Example 10.12 defined $e^A = \sum_{k\geq 0} A^k/k!$. Two facts are needed, both easy from the series: $e^A e^B = e^{A+B}$ when $AB = BA$, by multiplying the series and collecting terms as in the scalar case, and $\left(e^A\right)^t = e^{A^t}$, since transposition is continuous and linear.

Proposition 23.7 (Determinant of an exponential)**.** $\det e^A = e^{\operatorname{trace} A}$ *for every square* A.

Proof. By Theorem 14.5 write $A = UTU^*$ with T upper triangular carrying the eigenvalues λ_i. Then $e^A = Ue^TU^*$ term by term, and e^T is upper triangular with diagonal entries e^{λ_i}, since powers of an upper triangular matrix are upper triangular with the powers of the diagonal. So $\det e^A = \prod_i e^{\lambda_i} = e^{\sum_i \lambda_i} = e^{\operatorname{trace} A}$, the last step by Theorem 9.12. □



the axis is fixed and everything perpendicular to it turns

the spectrum: one real eigenvalue and a conjugate pair

Figure 23.2. Theorem 23.6. Composing two rotations of space about different axes gives a rotation about a third, which is not obvious and is exactly what the theorem asserts. The eigenvalues on the right are forced: they have modulus one because Q preserves lengths, they multiply to $\det Q = 1$, and the non-real ones come in conjugate pairs because Q is real, so in odd dimension one of them must be 1. That parity argument is why the theorem is about SO_3 and fails in SO_4, where a rotation can leave no axis fixed.

Theorem 23.8 (Skew-symmetric generates rotations)**.** *If $A^t = -A$ then $e^A \in SO_n$. Conversely every element of SO_n is e^A for some skew-symmetric A.*

Proof. For the first part, A commutes with $-A$, so

$$\left(e^A\right)^t e^A \;=\; e^{A^t} e^A \;=\; e^{-A} e^A \;=\; e^0 \;=\; \mathbf{1},$$

so e^A is orthogonal; and $\operatorname{trace} A = 0$ because the diagonal of a skew-symmetric matrix vanishes, so $\det e^A = e^0 = 1$ by Proposition 23.7. The converse we quote: it follows from the normal form for orthogonal matrices, which writes any $Q \in SO_n$ as a block diagonal matrix of plane rotations and identity blocks in a suitable orthonormal basis, each plane rotation being the exponential of a skew 2×2 block. □

Figure 23.3 is Lie's observation, and it is the reason a course in linear algebra is enough to say something about continuous symmetry: the skew-symmetric matrices form a vector space, and the exponential carries it onto the rotations.

Example 23.9 (Why angular velocity is a vector, and only in three dimensions)**.**

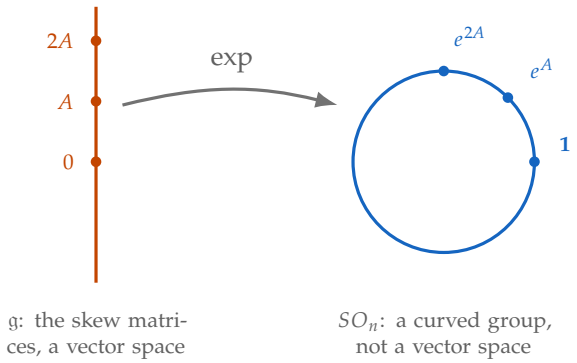


Figure 23.3. Theorem 23.8. This is Lie's observation, and it is the reason a course in linear algebra is enough to say something about continuous symmetry. The skew-symmetric matrices form a vector space of dimension $n(n-1)/2$, closed under the bracket $[A, B] = AB - BA$ but not under multiplication, and every question about rotations near the identity becomes a question about that vector space. For $n = 3$ its dimension is 3, which is why a rotation of space has three parameters and why angular velocity can be written as a vector at all.

For $w = (w_1, w_2, w_3)$ put

$$[w]_\times = \begin{pmatrix} 0 & -w_3 & w_2 \\ w_3 & 0 & -w_1 \\ -w_2 & w_1 & 0 \end{pmatrix},$$

which is skew-symmetric, and every skew-symmetric 3×3 matrix is of this form. A direct computation gives $[w]_\times v = w \times v$, the cross product of Definition 2.16. So $w \mapsto [w]_\times$ is an isomorphism from $\mathbb{R}^3$ to the skew-symmetric matrices, and $t \mapsto e^{t[w]_\times}$ is the rotation about the axis w at angular speed $\|w\|$. The coincidence that makes this possible is $n(n-1)/2 = n$, whose only solutions are $n = 3$ and the degenerate $n = 0$; at $n = 1$ the skew-symmetric matrices are the zero matrix alone, of dimension 0, against a one-dimensional $\mathbb{R}^1$, so the two do not match there either. In four dimensions the skew-symmetric matrices form a six-dimensional space and there is no angular velocity vector.

Example 23.10 (Rodrigues' formula)**.** *Let a be a unit vector and $K = [a]_\times$. Then $K^3 = -K$, which one checks directly, so the series for $e^{\theta K}$ collapses: separating the terms with even and odd powers and using $K^{2m+1} = (-1)^m K$ and $K^{2m+2} = (-1)^m K^2$,*

$$e^{\theta K} = \mathbf{1} + (\sin\theta)\, K + (1 - \cos\theta)\, K^2.$$

That is the rotation through θ about a, written without any trigonometry beyond two scalars, and it is what graphics and robotics code actually evaluates.

One caution about what the exponential reaches. Figure 23.1 is worth another look here: a matrix group can be disconnected, the determinant is a continuous function on it taking only the values ± 1, and the exponential of a tangent vector never leaves the component containing the identity.

23.4 Symmetry in learned models

Definition 23.11 (Invariance and equivariance)**.** *Let a group G act on the input space by matrices $\rho(g)$ and on the output space by matrices $\sigma(g)$. A map f is* invariant *if $f(\rho(g)x) = f(x)$ for all g, and* equivariant *if $f(\rho(g)x) = \sigma(g)f(x)$.*

Insight

Equivariance is a constraint on the weight matrices, and therefore a linear algebra question. Requiring $W\rho(g) = \sigma(g)W$ for every g says that W lies in the kernel of the linear map $W \mapsto W\rho(g) - \sigma(g)W$, an intersection of null spaces, which by Chapter 5 is a subspace. Building an equivariant layer means parameterising that subspace instead of all of $M_n(\mathbb{R})$, and the saving is the codimension. A convolution is the classical case: it is exactly a matrix commuting with all translations, which is why convolutional layers have so few parameters and why they were the first architecture to work.

Example 23.12 (Orthogonal weights and the conditioning problem)**.** *The singular values of Q are the square roots of the eigenvalues of Q^tQ, by Definition 16.2, so Q is orthogonal exactly when all of them equal 1; then $\|Q\|_2 = 1$ and, by Definition 16.10, $\kappa(Q) = 1$. An orthogonal matrix neither stretches nor shrinks in any direction. The spectral norm is submultiplicative, since $\|ABx\| \le \|A\|_2 \|Bx\| \le \|A\|_2 \|B\|_2 \|x\|$, and the smallest singular value is supermultiplicative by the same argument applied to inverses, so a stack of L layers whose weights all have singular values in $[s_{\min}, s_{\max}]$ scales a gradient by a factor between $s_{\min}^L$ and $s_{\max}^L$. That is the exploding and vanishing gradient problem, and it is geometric growth in L unless both bounds sit at 1. Constraining the weights to O_n, or to a neighbourhood of it, removes the problem by construction. Theorem 23.8 says how to do it with unconstrained parameters: store a general matrix A, use $Q = e^{(A-A^t)/2}$, and the constraint holds automatically because the exponential of a skew matrix is orthogonal.*

Caution

Orthogonal weights are not free. The constraint costs expressive power, it makes each step more expensive since the exponential must be evaluated or approximated, and the argument above is about the linear part only: the nonlinearity between the layers has singular values of its own and can undo the whole thing. As with the low-rank bet of Chapter 19, what the linear algebra supplies is a guarantee about one component, not about the system.

Remark 23.13 (Quaternions and SU_2)**.** *The group SU_2 consists of the matrices $\left(\begin{smallmatrix} \alpha & -\overline{\beta} \\ \beta & \overline{\alpha} \end{smallmatrix}\right)$ with $|\alpha|^2 + |\beta|^2 = 1$, so as a set it is the unit sphere in $\mathbb{C}^2 = \mathbb{R}^4$. Writing $\alpha = w + xi$ and $\beta = y + zi$ identifies it with the unit quaternions, and there is a surjective homomorphism $SU_2 \to SO_3$ whose kernel is $\{\pm\mathbf{1}\}$: two unit quaternions give the same rotation of space. That is why graphics and robotics store rotations as four numbers with one constraint rather than as three angles. Three angles are a chart on SO_3, and no chart covers it, which is what gimbal lock is.*

Open Problem 23.14 (When do two matrices generate a free group?)**.** *For a real number λ, let*

$$A_\lambda = \begin{pmatrix} 1 & \lambda \\ 0 & 1 \end{pmatrix}, \qquad B_\lambda = \begin{pmatrix} 1 & 0 \\ \lambda & 1 \end{pmatrix}.$$

For which λ is the group generated by A_λ and B_λ free of rank two, that is, subject to no relation except the ones forced by the group axioms?

What is known. *Sanov proved the group is free for $\lambda = 2$, and a ping-pong argument extends this to every λ with $|\lambda| \geq 2$. For λ transcendental it is free, by a specialisation argument. For rational $\lambda = p/q$ in $(0, 2)$ the picture is a mess: it is known to be non-free for $\lambda = 1$ and for various small rationals, and free for others, and* for infinitely many rationals in $(0, 2)$ the answer is unknown, *including values as simple as some with small denominators [439]. The set of $\lambda \in (0, 2)$ giving free groups is not known to be measurable.*

Where the difficulty sits. *Freeness means no nonempty reduced word in A and B equals the identity, which is an infinite family of polynomial conditions on λ. Each individual word gives a polynomial equation, so the non-free λ form a countable union of finite sets and are therefore countable; freeness holds off a countable set.* But deciding membership for a specific λ requires ruling out every word at once, *and the ping-pong argument, which is the only general tool, stops working below $|\lambda| = 2$ because the two ping-pong*

regions start to overlap.
Where a reader might start. *Run the ping-pong argument for $\lambda = 2$: let X be the vectors with $|x_1| > |x_2|$ and Y its complement, and check that nonzero powers of A_2 carry Y into X and powers of B_2 carry X into Y. Then set $\lambda = 1.9$ and find exactly where the two regions stop separating. That failure point is the frontier.*

Exercises

Exercise 23.15. *Verify from Definition 23.1 that $SL_n(\mathbb{R})$ and the invertible upper triangular matrices are groups, and that the symmetric invertible matrices are not. Which axiom fails in the last case, and give a 2×2 counterexample.*

Exercise 23.16. *Work out G_β explicitly for $n = 2$ and $M = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, using Proposition 23.3. Show that the connected piece containing $\mathbf{1}$ consists of the matrices $\left(\begin{smallmatrix} \cosh t & \sinh t \\ \sinh t & \cosh t \end{smallmatrix}\right)$, and say which quantity these preserve in place of length.*

Exercise 23.17. *Write down the reflection in the line at angle φ to the x-axis as a 2×2 matrix, check that it is orthogonal with determinant -1, and verify by direct multiplication that the composition of the reflections in the lines at angles φ and ψ is the rotation $R_{2(\varphi-\psi)}$.*

Exercise 23.18. *Let Q be the rotation taking $e_1 \mapsto e_2 \mapsto e_3 \mapsto e_1$. Confirm that $Q \in SO_3$, find its axis by solving $Qa = a$, and find its angle from $\operatorname{trace} Q = 1 + 2\cos\theta$. Why is that trace formula true?*

Exercise 23.19. *Verify Proposition 23.7 by hand for $A = \left(\begin{smallmatrix} 0 & -\theta \\ \theta & 0 \end{smallmatrix}\right)$, by computing e^A in closed form and taking its determinant. Which element of SO_2 is it?*

Exercise 23.20. *Show that the skew-symmetric $n \times n$ matrices form a vector space of dimension $n(n-1)/2$ by exhibiting a basis. Confirm that the bracket $[A, B] = AB - BA$ of two skew-symmetric matrices is skew-symmetric, and that the ordinary product need not be.*

Exercise 23.21. *Check that $[w]_\times v = w \times v$ for $w = (1,2,3)$ and $v = (0,1,-1)$, and verify $K^3 = -K$ for $K = [a]_\times$ with a a unit vector. Then use Example 23.10 to write the rotation through $\pi/2$ about $(0,0,1)$ and check it against the matrix you would have written down directly.*

Exercise 23.22. *Let G be the cyclic group generated by the cyclic shift P on $\mathbb{R}^3$. Find every $W \in M_3(\mathbb{R})$ with $WP = PW$, check that the answer is a subspace as the Insight predicts, and give its dimension. What are such matrices called, and what does the dimension say about the parameter count of a convolution against that of a general layer?*

Exercise 23.23. *Take $A = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ and compute $Q = e^{tA}$ for $t = \pi/4$. Confirm $\kappa(Q) = 1$ using the singular values. Then explain in two sentences why Example 23.12 must skew-symmetrise a general stored matrix rather than exponentiating it as it stands, and check on $A = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 2 \end{smallmatrix}\right)$ that e^A need not be orthogonal.*

References and further reading

Stillwell [587] is the gentlest serious route from this chapter to Lie theory and does almost everything with matrices, which is the right level for a reader arriving from here; Hall's book [293] is the standard next step and proves the surjectivity quoted in Theorem 23.8 along with the normal form for orthogonal matrices. Artin [28] develops the classical groups from the forms of Chapter 22 exactly as Proposition 23.3 suggests, and is the best short account of the symplectic case. Weyl [642] is the book that named them and is still worth opening, though not first. Banica [51], our baseline, devotes its final part to matrix groups, including the finite and compact cases and the random matrix theory that grows out of them, and this chapter is the elementary entrance to it. For the group-theoretic view of architecture in machine learning the literature is young and moving quickly, and the reader is better served by the current papers than by any textbook account.